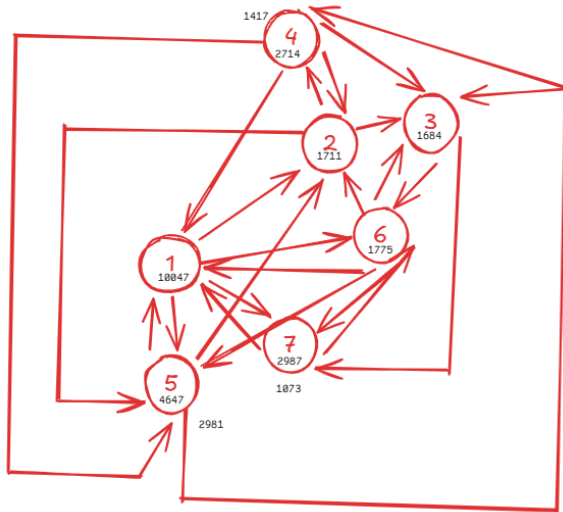


Module 10 – MOLP

Exploratory Data Analysis



Model Formulation

Decision Variables

X_{ij} = amount of goods shipped from node i to node j

$$X_{12} = 1711$$

$$X_{15} = 4647$$

$$X_{17} = 2987$$

$$X_{53} = 701$$

$$X_{46} = 1417$$

$$X_{64} = 266$$

$$X_{71} = 2714$$

$$X_{76} = 1073$$

Objective Function

$$\text{Min Transportation Cost} = 11X_{12} + 7X_{15} + 18X_{17} + 17.7X_{46} + 9X_{53} + 13X_{64} + 12X_{71} + 24X_{76}$$

$$\text{Min Distance} = 17.4X_{12} + 3.5X_{15} + 31.2X_{17} + 6X_{46} + 9.3X_{53} + 12.3X_{64} + 31.2X_{71} + 2.2X_{76}$$

$$\text{Min Non-Eco} = 0X_{12} + 1X_{15} + 0X_{17} + 1X_{46} + 1X_{53} + 0X_{64} + 1X_{71} + 0X_{76}$$

$$\text{Min Congestion} = 95X_{12} + 20X_{15} + 34X_{17} + 84X_{46} + 88X_{53} + 92X_{64} + 87X_{71} + 30X_{76}$$

Constraints

$$TC - 123851 / 132851 \leq Z$$

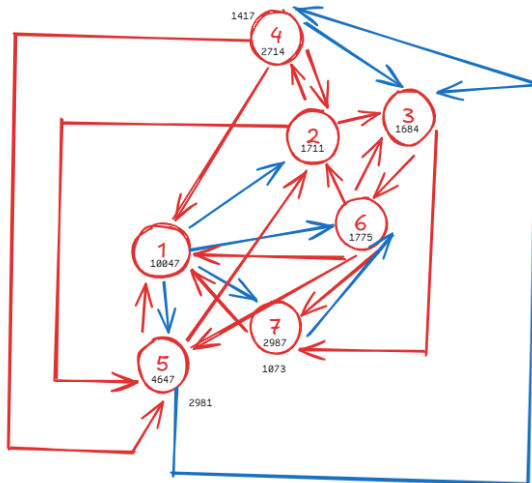
$$D - 87081 / 87081 \leq Z$$

$$NE - 11344 / 11344 \leq Z$$

$$2 * C - 3395 / 3395 \leq Z$$

Model Optimized for Equally Weighted Objectives

Ship	From	To	Lat from	Long from	Lat to	Long to	Unit Cost	Distance	Transportation Method	Eco Friendly	non eco	Congestion	High/Low		
17111	1	Butter Pecan Buns	37.5°	-102.5°	31.68°	-118.85°	11	17.4	Electrified Rail	0	0	1	95	1	
4647	1	Popping Candy Plains	37.5°	-102.5°	38.86°	-101.36°	7	3.5	Air Freight	0	0	1	20	0	
701.365	1	Butter Pecan Buns	6	Pudding Peaks	37.5°	-99.31°	18	3.4	Cargo Ships (Heavy Fuel Oil)	0	1	1	78	1	
2987.5	1	Butter Pecan Buns	37.5°	-102.5°	7.1	101.36°	19	18.0	Cargo Ships (Heavy Fuel Oil)	0	1	1	34	0	
0	2	Fruity Oushier Geyser	31.68°	-118.85°	44.19°	-101.34°	17	21.5	Wind-powered Ships	1	0	0	102	1	
0	2	Fruity Oushier Geyser	4	Jellybean Jungle	31.68°	-118.85°	38.86°	-114.75°	12	8.3	Air Freight	0	1	24	0
0	2	Fruity Oushier Geyser	5	Popping Candy Plains	31.68°	-118.85°	34.2°	-101.36°	6	17.7	Air Freight	0	1	27	0
0	3	Gingerbread Glades	44.19°	-101.34°	38.86°	-99.31°	11	6.0	Cargo Ships (Heavy Fuel Oil)	0	1	1	28	0	
0	3	Gingerbread Glades	6	Gingerbread Glades	31.68°	-114.75°	34	11	3.0	Steamshiping Cargo Ships	0	1	70	0	
0	4	Jellybean Jungle	1	Butter Pecan Buns	38.86°	-114.75°	37.5°	-102.5°	23	12.3	Air Freight	0	1	93	1
0	4	Jellybean Jungle	2	Fruity Oushier Geyser	38.86°	-114.75°	31.68°	-118.85°	11	8.3	Slow Steaming Cargo Ships	0	1	93	1
1417.53	4	Jellybean Jungle	3	Gingerbread Glades	38.86°	-114.75°	44.19°	-101.34°	6	14.4	Diesel Rail	0	1	103	1
0	4	Jellybean Jungle	5	Popping Candy Plains	38.86°	-114.75°	34.2°	-101.36°	6	17.2	Cargo Ships (Heavy Fuel Oil)	0	1	98	1
0	5	Popping Candy Plains	34.2°	-101.36°	1	Butter Pecan Buns	37.5°	-102.5°	12	3.5	Air Freight	0	1	97	1
0	5	Popping Candy Plains	2	Fruity Oushier Geyser	34.2°	-101.36°	31.68°	-118.85°	16	17.7	Diesel Rail	0	1	97	1
0	5	Popping Candy Plains	3	Gingerbread Glades	34.2°	-101.36°	44.19°	-101.34°	15	10.0	Diesel Trucks	0	1	85	1
266.465	5	Popping Candy Plains	4	Jellybean Jungle	34.2°	-101.36°	38.86°	-114.75°	7	14.2	Diesel Trucks	0	1	84	1
2714.53	5	Popping Candy Plains	1	Butter Pecan Buns	34.2°	-101.36°	37.5°	-102.5°	5	3.4	Electric Rail	1	0	97	1
0	6	Pudding Peaks	38.86°	-99.31°	31.68°	-118.85°	22	20.7	Cargo Ships (Heavy Fuel Oil)	0	1	1	84	1	
0	6	Pudding Peaks	3	Gingerbread Glades	38.86°	-99.31°	44.19°	-101.34°	10	6.0	Electric/Hybrid Trucks	1	0	92	1
0	6	Pudding Peaks	5	Popping Candy Plains	38.86°	-99.31°	34.2°	-101.36°	18	4.8	Diesel Trucks	0	1	33	0
0	6	Pudding Peaks	7	Turkish Delight Tundra	38.86°	-99.31°	40.45°	-100.39°	11	2.2	Diesel Rail	0	1	34	0
0	7	Turkish Delight Tundra	40.45°	-100.39°	37.5°	-102.5°	18	2.6	Diesel Trucks	0	1	87	0		
1073.63	7	Turkish Delight Tundra	6	Pudding Peaks	40.45°	-100.39°	38.86°	-99.31°	24	2.2	Air Freight	0	1	30	0



My model is a multi-objective network flow optimization framework designed to determine the most efficient way to ship goods from a central supply node (Butter Pecan Bluff) to multiple demand nodes across a transportation network. It considers several shipping routes, each with associated costs, distances, environmental impacts, and congestion levels, using various transportation modes like trucks, ships, and air freight. The model's goal is to balance four competing objectives—minimizing total transportation cost, travel distance, use of non-eco-friendly transport, and congestion—by minimizing the maximum weighted percentage deviation from target values for each objective. Through flow constraints and decision variables representing shipment amounts between nodes, the model ensures supply meets demand while optimizing performance across all criteria.

Model with Stipulation

I selected a balanced configuration that places the highest emphasis on congestion (weight = 3), moderate emphasis on cost and eco-friendliness (1.5 each), and less emphasis on distance (1). This reflects a real-world logistics priority where congestion, especially in urban centers—can severely impact delivery times and environmental goals. While cost and sustainability remain key business concerns, marginal gains in route length are less impactful than avoiding bottlenecks or emissions-heavy routes. This setup produced a routing strategy that reduced congestion and increased the use of cleaner transport options while maintaining reasonable cost levels, aligning well with both operational efficiency and corporate sustainability goals.